Evaluating the Transport Impact of Large Post-Quantum DNSSEC Signatures

Theoretical Part (Project – Part 1)

Justin Kasper
Student no. 145971
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Mert Duman
Student no. 145872
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Efim Shchegolev
Student no. 144376
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Sezer Tetik
Student no. 145873
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

***Abstract*—Post-quantum signature schemes such as ML-DSA-44, standardized by NIST in FIPS 204, produce signatures and public keys that are one to two orders of magnitude larger than those of the classical algorithms used in DNSSEC today. As a result, signed DNS responses routinely exceed the 1,232-byte EDNS(0) payload recommendation and the typical Internet MTU, which pushes them onto IP fragmentation, truncation with TCP fallback, or alternative transports. This theoretical part reviews the DNS and DNSSEC foundations, the classical and post-quantum signature algorithms relevant to DNSSEC, and the message-size and transport mechanisms that govern how large responses are delivered, including TCP fallback, Application-layer Request-based Resource Record Fragmentation (ARRF), and DNS over QUIC (DoQ). It then surveys related work and identifies the gap addressed by the practical part of the project: a controlled, reproducible comparison of these transports for ML-DSA-44 against an ECDSA baseline under varying latency and packet loss.**

***Index Terms*—DNSSEC, post-quantum cryptography, ML-DSA, TCP fallback, IP fragmentation, ARRF, DNS over QUIC**

I. Introduction

The Domain Name System (DNS) translates human-readable domain names into the numerical addresses that are required for almost every transaction on the Internet [1]. Despite this critical role, the original DNS protocol was not designed with any mechanism for verifying that a response actually originates from the authoritative source or that it has not been altered in transit [2]. The DNS Security Extensions (DNSSEC) were introduced to close this gap by adding data origin authentication and data integrity protection to DNS responses through digital signatures, without, however, providing confidentiality [2]–[4]. More than two decades after its initial specification, DNSSEC is considered the best current practice for authenticating DNS data [5], even though deployment remains uneven: recent estimates suggest that fewer than ten percent of website domain names are signed and that only around a third of queries to recursive resolvers are actually validated [5].

The cryptographic foundation of DNSSEC, however, is now under long-term threat. The signature algorithms that are most widely deployed today, in particular RSA and the Elliptic Curve Digital Signature Algorithm (ECDSA) [6], [7], derive their security from the computational hardness of integer factorization and the discrete logarithm problem. A sufficiently large, cryptographically relevant quantum computer would be able to solve both of these problems efficiently using Shor's algorithm [8], which would allow an attacker to forge DNSSEC signatures and therefore to present spoofed DNS data as authentic.[1] To address this risk, the U.S. National Institute of Standards and Technology (NIST) has standardized the Module-Lattice-Based Digital Signature Algorithm (ML-DSA) in FIPS 204, a signature scheme whose security relies on the hardness of lattice problems and that is believed to remain secure even against an adversary equipped with a quantum computer [9].

Post-quantum cryptography (PQC) schemes such as ML-DSA, however, do not come without cost. The smallest standardized ML-DSA parameter set, ML-DSA-44, produces signatures of 2,420 octets and public keys of 1,312 octets [9], [10], which is one to two orders of magnitude larger than a typical ECDSA signature. For comparison, an ECDSA P-256 signature occupies only 64 bytes, so an ML-DSA-44 signature is almost 38 times larger [11]. When such a signature is embedded into a DNSSEC response together with the DNS

Network and Information Systems Security, ISCTE – Instituto Universitário de Lisboa. Submitted on 28 October 2026.

[1]This is a long-term, not an immediate, threat: at the time of writing, no cryptographically relevant quantum computer is publicly known to exist. Nevertheless, migration planning for long-lived infrastructure such as DNS is widely recommended well in advance of such a scenario.

header, resource records, and—where applicable—a DNSKEY record, the resulting message can easily exceed the conservative 1,232-byte UDP payload size that is commonly recommended in operational practice today, and it will frequently exceed the roughly 1,500-byte Maximum Transmission Unit (MTU) of a typical Ethernet path [12], [13]. Because DNS was originally designed around a 512-byte UDP message size [1] and later extended via EDNS(0) to negotiate larger UDP payloads [12], oversized post-quantum DNSSEC responses risk falling back to one of two imperfect mechanisms: IP-layer fragmentation, which is unreliable because many firewalls and other middleboxes silently drop fragmented packets [14], [15], or truncation of the UDP response followed by a retry over TCP, which guarantees delivery but adds at least one additional round trip and extra server-side state [14]. A drop-in replacement of classical signatures with quantum-safe ones is expected to increase DNSSEC query resolution times by at least a factor of two: a response carrying large post-quantum signatures is likely to be truncated, which wastes the initial UDP round trip, and the subsequent retry over TCP incurs at least two further round trips because of the three-way handshake [16].

The DNS operator community has already begun to experiment with mitigations for this problem. Cloudflare has deployed experimental ML-DSA-44 validation on its public 1.1.1.1 resolver, relying on TCP fallback whenever a response becomes oversized [11], and an open-source extension of the CoreDNS authoritative server has demonstrated on-the-fly signing with several PQC signature families, including ML-DSA [17], [18]. At the same time, the research literature has proposed alternative transport mechanisms that try to avoid the drawbacks of both IP fragmentation and plain TCP fallback. Application-layer Request-based Resource Record Fragmentation (ARRF) fragments oversized DNS resource records at the DNS protocol layer itself rather than relying on the IP layer, with the resolver requesting any missing fragments explicitly [19]. DNS over QUIC (DoQ) instead moves the entire DNS exchange onto a dedicated, encrypted QUIC connection, which has its own packetization and loss-recovery behavior and therefore sidesteps classical IP fragmentation entirely [20].

Although each of these mechanisms—plain TCP fallback, ARRF, and DoQ—has been described or prototyped individually, a direct, controlled, and reproducible comparison of their latency, bandwidth, and reliability behavior specifically under the conditions created by post-quantum DNSSEC signatures is still largely missing from the public literature. This gap motivates the present project. Using ECDSA-based DNSSEC as a classical baseline and an ML-DSA-44-based DNSSEC deployment as the post-quantum case, this project evaluates DNS resolution under simulated network conditions with varying latency and packet loss, measuring query latency, resource utilization, bandwidth consumption, packet sizes, and the frequency of TCP fallback. Time permitting, the evaluation is extended to ARRF-based fragmentation and to DNS over QUIC, in order to assess which combination of signature algorithm and transport mechanism currently represents the most realistic path toward a quantum-resistant DNSSEC deployment.

Concretely, the project is guided by the following research questions, which will be answered in the practical part (Project – Part 2):

RQ1: How much do ML-DSA-44 signatures increase DNSSEC response sizes, packet sizes, and bandwidth consumption, and how often do they trigger truncation and TCP fallback, compared with the ECDSA baseline?

RQ2: How do the end-to-end query latency and resource utilization of post-quantum DNSSEC with TCP fallback change as path latency and packet loss increase?

RQ3: (Time permitting.) To what extent can ARRF or DNS over QUIC reduce the overhead identified in RQ1 and RQ2?

The remainder of this (theoretical) part of the report is structured as follows. Section II introduces the theoretical foundations that are necessary to understand the problem: the DNS and DNSSEC protocols, the classical and post-quantum signature algorithms relevant to this project, and the mechanisms that govern DNS message size and transport selection. Section **??** positions this project with respect to the existing state of the art and related work. Section **??** closes the theoretical part with a brief summary and a transition toward the practical implementation and evaluation that are reported in Part 2 of this project.

II. Theoretical Foundations

A. The Domain Name System and Its Security Model

The Domain Name System provides a globally distributed, hierarchical naming service that maps domain names to resource records, most commonly IP addresses [1]. A DNS resolver issues a query for a given name and record type and receives a response from an authoritative name server, typically via a recursive resolver that caches intermediate results. In its original specification, DNS offers no cryptographic guarantee that a received response was actually produced by the zone's authoritative server or that it was not modified or injected by an on-path or off-path attacker [2]. This absence of origin authentication is the central weakness that DNSSEC was designed to address.

B. DNS Security Extensions (DNSSEC)

1) Core Resource Records and the Chain of Trust: DNSSEC, as specified across RFC 4033, RFC 4034, and RFC 4035, extends DNS with a small set of new resource record types and protocol behaviors that together provide data origin authentication and data integrity [2]–[4]. A zone owner signs its resource record sets (RRsets) with a private key, producing RRSIG records that contain the resulting digital signature; the corresponding public key is published in a DNSKEY record. A DS (Delegation Signer) record, stored in the parent zone, contains a cryptographic hash of the child zone's DNSKEY and thereby links the child's key material into the chain of trust that ultimately terminates at the signed

DNS root. A validating resolver can therefore follow this chain from the root down to the queried name, verifying each signature along the way, to establish that a response is authentic. Importantly, DNSSEC does not encrypt DNS traffic and therefore provides no confidentiality; its guarantees are limited to authenticity and integrity [2].

RFC 4033 additionally distinguishes two roles for zone keys: a Key Signing Key (KSK) corresponds to a private key used to sign one or more other authentication keys of a zone, whereas a Zone Signing Key (ZSK) corresponds to a private key used to sign the zone itself; typically, the KSK signs the DNSKEY RRset containing the ZSK, and the ZSK signs the other zone data [2]. This distinction is purely operational: validation does not differentiate between the two roles, and a single key may serve as both [2]. The DS RRset in the parent zone indicates the key(s) used to self-sign the child's DNSKEY RRset, so the typical authentication chain is DNSKEY $\rightarrow$ [DS $\rightarrow$ DNSKEY]* $\rightarrow$ RRset [2]. Fig. 1 illustrates how DS, DNSKEY, and RRSIG records combine into the chain of trust [2]–[4]. Two properties of this design are directly relevant for this project. First, when a query has the DO bit set, a security-aware authoritative server should send the RRSIG records needed to authenticate the RRsets in the response and keep each RRset together with its RRSIG(s) [4], while the DNSKEY RRset carries the complete public keys of the zone [3]; the sizes of signatures and keys therefore translate directly into the size of DNS messages. Second, a sender must not use DNS name compression on the Signer's Name field of an RRSIG record, so this field always contributes its full length [3].

2) Adoption and Current Deployment Status: Although DNSSEC is regarded as the current best practice for DNS origin authentication, and although the DNS root zone and many top-level domains (TLDs) are signed, end-to-end deployment below the TLD level remains limited: current estimates indicate that fewer than ten percent of website domain names are DNSSEC-signed and that only roughly a third of queries handled by recursive resolvers are actually validated [5]. This uneven deployment is relevant to the present project because any additional transport overhead introduced by post-quantum signatures could further discourage adoption if it is not kept within acceptable latency and reliability bounds. Support for TCP, the fallback transport, is also not universal: according to measurements published in 2016 and cited by the authors of [21], about 11% of name servers lacked TCP support [21].

C. Cryptographic Signature Algorithms and the Quantum Threat

1) Classical Signature Algorithms Used in DNSSEC: The DNSSEC specifications allow multiple signature algorithms to be registered and used, and operational guidance on which algorithms should be implemented and used for signing or validation is maintained separately from the core protocol documents [7]. In current practice, RSA and ECDSA are the most widely deployed choices [6], [7]. ECDSA in particular is attractive for DNS because it produces comparatively small signatures and public keys, which keeps DNSSEC-signed responses close to the message sizes that classical, unsigned DNS traffic already uses. For this reason, ECDSA is used as the classical baseline algorithm in this project.

Concretely, an ECDSA P-256 (DNSSEC algorithm 13) signature is encoded as the concatenation $r \,|\, s$ of two 32-octet integers, i.e., 64 octets [6], and the public key is the uncompressed curve point $x \,|\, y$, which occupies 64 bytes [6], [11]; RSA-2048 with SHA-256 (algorithm 8) uses 256-byte signatures and 260-byte public keys [11]. The Edwards-curve algorithm Ed25519 is even more compact, with 32-octet public keys and 64-octet signatures [22]. All of these values remain far below the 1,232-byte UDP payload limit discussed in Section **??**.

2) The Quantum Threat to Classical Public-Key Cryptography: Both RSA and ECDSA rely on mathematical problems—integer factorization and the elliptic-curve discrete logarithm problem, respectively—that are believed to be intractable for classical computers at the key sizes used in practice. A sufficiently large quantum computer running Shor's algorithm [8] would, however, be able to solve both problems in polynomial time. If such a quantum computer became available, an attacker could use it to derive the private signing key of a DNSSEC zone (or otherwise forge valid-looking signatures) from publicly available information, defeating the authentication guarantees that DNSSEC is meant to provide. Because DNS is expected to remain a core piece of Internet infrastructure for the foreseeable future, and because there is currently no widely deployed alternative to it, this motivates investigating post-quantum alternatives for DNSSEC signing well ahead of the point at which such an attack becomes practical [9].

3) Post-Quantum Cryptography and the ML-DSA Standard: In August 2024, NIST finalized FIPS 204, the Module-Lattice-Based Digital Signature Standard (ML-DSA), as one of the first standardized post-quantum digital signature algorithms [9]. ML-DSA's security is based on the conjectured hardness of problems over module lattices, specifically variants of the Module Learning With Errors and Module Short Integer Solution problems, for which no efficient quantum algorithm is currently known. FIPS 204 defines three parameter sets, ML-DSA-44, ML-DSA-65, and ML-DSA-87, offering increasing security levels at increasing signature and key sizes [9]. An Internet-Draft from the DNS Operations working group specifies how ML-DSA-44 can concretely be carried in DNSSEC's DNSKEY and RRSIG resource records, and states that an ML-DSA-44 public key occupies 1,312 octets while a corresponding signature occupies 2,420 octets [10]. ML-DSA-44 is chosen in that document, and consequently also in this project, because it offers the smallest keys and signatures among the three ML-DSA parameter sets while still targeting NIST security category 2, making it the most DNS-friendly of the standardized options [10]. This Internet-Draft is, at the time of writing, still a work in progress and has not yet been published as an RFC; it is used here as the most concrete and current specification of how ML-DSA integrates with DNSSEC, but

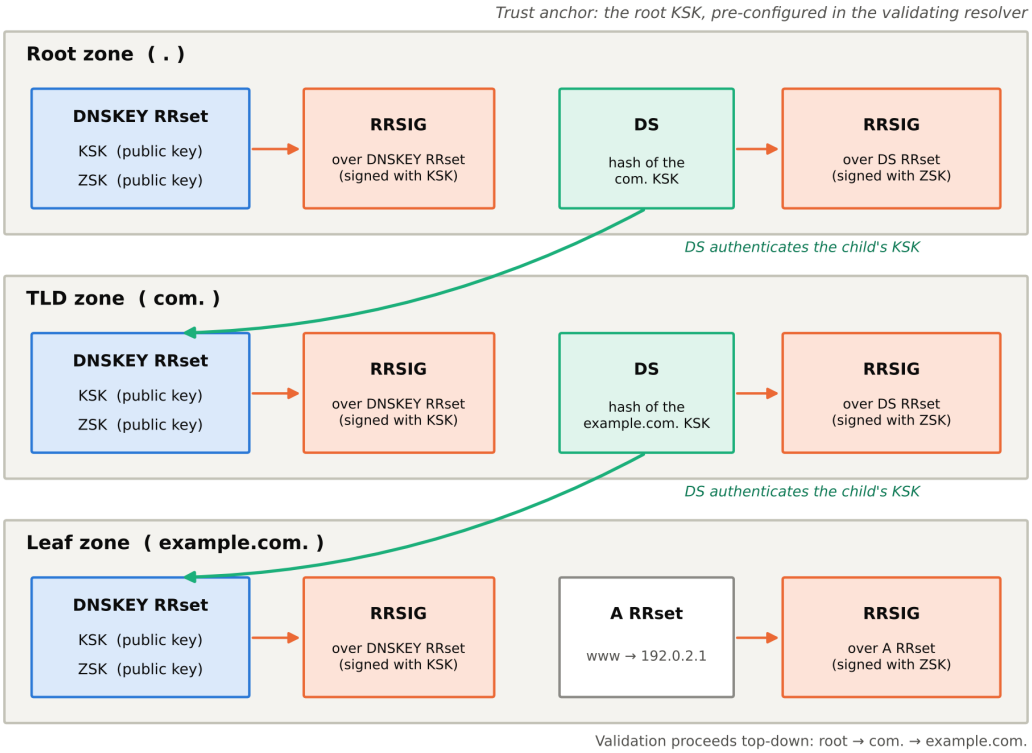


Fig. 1. Simplified DNSSEC chain of trust with the typical KSK/ZSK split.

its details may still change before final standardization.

Table I and Fig. 2 place ML-DSA-44 in the context of other classical and post-quantum signature algorithms; the sizes are taken from [6], [9]–[11], [17], [22], and the thresholds marked in Fig. 2 are the 1,232-byte EDNS(0) recommendation [23] and the ≈1,500-byte Internet MTU [13]. FIPS 204 states that ML-DSA is derived from CRYSTALS-Dilithium, one of the schemes selected in the NIST post-quantum standardization process [9]. Among the post-quantum schemes integrated into CoreDNS in [17], Falcon-512 has a considerably smaller signature (752 bytes) than ML-DSA-44 (2,420 bytes), while SPHINCS+-128s combines a very small 32-byte public key with a 7,856-byte signature [17]. The same study reports that all evaluated post-quantum schemes exceed typical UDP size limits and therefore force the use of TCP, whereas the classical algorithms stay within these limits [17].

The authors of [21] formulated explicit requirements for DNSSEC signature algorithms, the most important being that a signature should not exceed 1,232 bytes; further requirements are a validation speed of at least 1,000 signatures per second and a signing speed of at least 100 signatures per second; public keys of up to 64 kB are considered good, whereas larger keys are accepted only conditionally [21]. Every ML-DSA parameter set already violates the first requirement on signature size alone, which motivates the focus of this project on how the resulting large responses are transported rather than on the choice of signature algorithm alone.

TABLE I
PUBLIC-KEY AND SIGNATURE SIZES OF SIGNATURE ALGORITHMS RELEVANT TO DNSSEC

Algorithm	Family	Public key (B)	Signature (B)
Ed25519 [22]	Classical (elliptic curve)	32	64
ECDSA P-256 [6], [11]	Classical (elliptic curve)	64	64
RSA-2048/SHA-256 [11]	Classical (RSA)	260	256
Falcon-512 [17]	Post-quantum (lattice)	897	752 (666 padded)
ML-DSA-44 [9], [10], [17]	Post-quantum (lattice)	1,312	2,420
ML-DSA-65 [17]	Post-quantum (lattice)	1,952	3,309
ML-DSA-87 [17]	Post-quantum (lattice)	2,592	4,627
SPHINCS+-SHA2-128s [17]	Post-quantum (hash-based)	32	7,856

REFERENCES

[1] P. Mockapetris, "Domain names—Implementation and specification," RFC 1035, STD 13, Internet Engineering Task Force (IETF), Nov. 1987. [Online]. Available: https://www.rfc-editor.org/rfc/rfc1035. [Accessed: Oct. 20, 2026].

[2] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "DNS security introduction and requirements," RFC 4033, Internet Engineering Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4033. [Accessed: Oct. 20, 2026].

[3] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "Resource records for the DNS security extensions," RFC 4034, Internet Engineer-

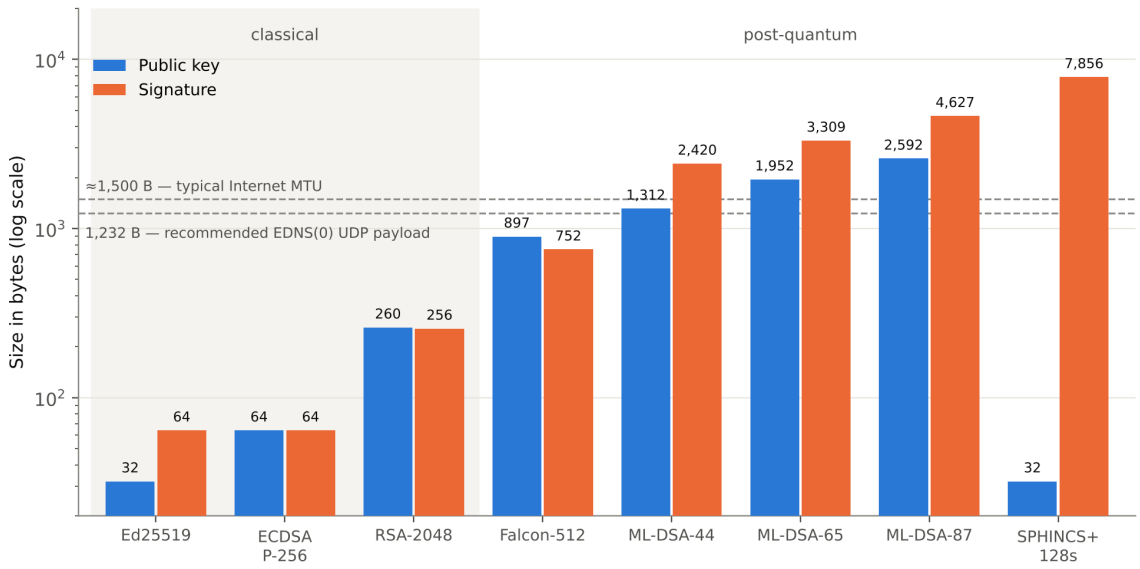


Fig. 2. Public-key and signature sizes (logarithmic scale) compared with the 1,232-byte EDNS(0) recommendation and the ≈1,500-byte Internet MTU.

ing Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4034. [Accessed: Oct. 20, 2026].

[4] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "Protocol modifications for the DNS security extensions," RFC 4035, Internet Engineering Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4035. [Accessed: Oct. 20, 2026].

[5] P. Hoffman, "DNS security extensions (DNSSEC)," RFC 9364, BCP 237, Internet Engineering Task Force (IETF), Feb. 2023. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9364. [Accessed: Oct. 20, 2026].

[6] P. Hoffman and W. C. A. Wijngaards, "Elliptic curve digital signature algorithm (DSA) for DNSSEC," RFC 6605, Internet Engineering Task Force (IETF), Apr. 2012. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6605. [Accessed: Oct. 20, 2026].

[7] P. Wouters and O. Sury, "Algorithm implementation requirements and usage guidance for DNSSEC," RFC 8624, Internet Engineering Task Force (IETF), Jun. 2019. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8624. [Accessed: Oct. 20, 2026].

[8] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM J. Comput.*, vol. 26, no. 5, pp. 1484–1509, 1997, doi: 10.1137/S0097539795293172.

[9] National Institute of Standards and Technology, *Module-Lattice-Based Digital Signature Standard*, FIPS 204, Aug. 2024, doi: 10.6028/NIST.FIPS.204. [Online]. Available: https://csrc.nist.gov/pubs/fips/204/final. [Accessed: Oct. 20, 2026].

[10] B. E. Westerbaan and S. Schmieg, "Module-lattice digital signature algorithm for DNSSEC," IETF Internet-Draft draft-westerbaan-dnssec-mldsa-04 (work in progress), Aug. 2026. [Online]. Available: https://datatracker.ietf.org/doc/draft-westerbaan-dnssec-mldsa/. [Accessed: Oct. 20, 2026].

[11] S. Neuteboom and B. Westerbaan, "1.1.1.1 now supports post-quantum DNSSEC, all 2,420 bytes of it," Cloudflare Blog, Sep. 10, 2026. [Online]. Available: https://blog.cloudflare.com/post-quantum-dnssec-1111/. [Accessed: Oct. 2, 2026].

[12] J. Damas, M. Graff, and P. Vixie, "Extension mechanisms for DNS (EDNS(0))," RFC 6891, STD 75, Internet Engineering Task Force (IETF), Apr. 2013. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6891. [Accessed: Oct. 20, 2026].

[13] Cloudflare, "Maximum transmission unit and maximum segment size," Cloudflare WAN Documentation, Aug. 24, 2026. [Online]. Available: https://developers.cloudflare.com/cloudflare-wan/reference/mtu-mss/. [Accessed: Oct. 2, 2026].

[14] J. Dickinson, S. Dickinson, R. Bellis, A. Mankin, and D. Wessels, "DNS transport over TCP—Implementation requirements," RFC 7766, Internet Engineering Task Force (IETF), Mar. 2016. [Online]. Available: https://www.rfc-editor.org/rfc/rfc7766. [Accessed: Oct. 20, 2026].

[15] A. Herzberg and H. Shulman, "Fragmentation considered poisonous, or: One-domain-to-rule-them-all.org," in *Proc. 2013 IEEE Conf. Communications and Network Security (CNS)*, 2013, pp. 224–232, doi: 10.1109/CNS.2013.6682711.

[16] A. S. Rawat and M. P. Jhanwar, "Post-quantum DNSSEC with faster TCP fallbacks," in *Progress in Cryptology – INDOCRYPT 2024*, Lecture Notes in Computer Science. Cham, Switzerland: Springer, 2024, pp. 212–236, doi: 10.1007/978-3-031-80311-6_11.

[17] J. Gento Suela, J. Blanco-Romero, F. Almenares Mendoza, and D. Díaz-Sánchez, "Implementing and evaluating post-quantum DNSSEC in CoreDNS," in *Proc. 2025 Int. Conf. Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, 2025, pp. 420–427, doi: 10.1109/MSWIM67937.2025.11308970.

[18] CoreDNS, "CoreDNS," GitHub repository. [Online]. Available: https://github.com/coredns/coredns. [Accessed: Oct. 2, 2026].

[19] J. Goertzen and D. Stebila, "Post-quantum signatures in DNSSEC via request-based fragmentation," in *Proc. 14th Int. Conf. Post-Quantum Cryptography (PQCrypto 2023)*, Lecture Notes in Computer Science, vol. 14154, 2023, pp. 535–564, doi: 10.1007/978-3-031-40003-2_20.

[20] C. Huitema, S. Dickinson, and A. Mankin, "DNS over dedicated QUIC connections," RFC 9250, Internet Engineering Task Force (IETF), May 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9250. [Accessed: Oct. 2, 2026].

[21] M. Müller, J. de Jong, M. van Heesch, B. Overeinder, and R. van Rijswijk-Deij, "Retrofitting post-quantum cryptography in Internet protocols: A case study of DNSSEC," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 50, no. 4, pp. 49–57, Oct. 2020, doi: 10.1145/3431832.3431838.

[22] O. Sury and R. Edmonds, "Edwards-curve digital security algorithm (EdDSA) for DNSSEC," RFC 8080, Internet Engineering Task Force (IETF), Feb. 2017. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8080. [Accessed: Oct. 29, 2026].

[23] K. Fujiwara and P. Vixie, "IP fragmentation avoidance in DNS over UDP," RFC 9715, Internet Engineering Task Force (IETF), Jan. 2025. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9715. [Accessed: Oct. 29, 2026].